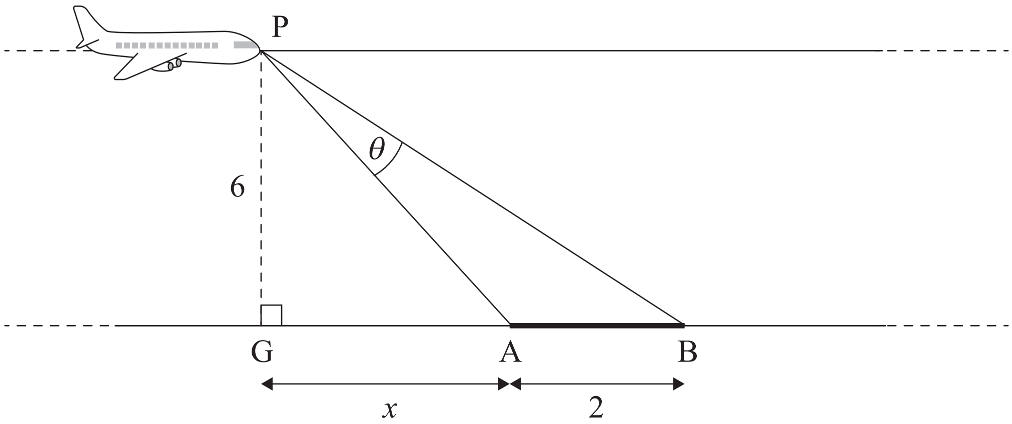
9. [Maximum mark: 9]

An airplane, P, is flying over horizontal ground at a constant height of $6\,\mathrm{km}$ and travelling at a constant speed. It is approaching a runway, $[\mathrm{AB}]$, which is $2\,\mathrm{km}$ in length.

Let G be the point on the ground directly below the airplane. When $\mathrm{GA} = x\,\mathrm{km}$, the pilot's viewing angle of the runway, $\mathrm{A\hat{P}B}$, is θ.

This is shown in the following diagram.

diagram not to scale



(a) Show that $\theta = \arctan\left(\dfrac{x+2}{6}\right) - \arctan\left(\dfrac{x}{6}\right)$. [2]

When the viewing angle is 0.178 radians, the rate at which the viewing angle is changing is 12.5 radians per hour.

(b) Find the speed of the airplane. [7]